

SRI CHAITANYA IIT ACADEMY, INDIA

AP • TELANGANA • KARNATAKA • TAMILNADU • MAHARASHTRA • DELHI • RANCHI

**PRAGYA G8 – ACHIEVERS | MATHEMATICS
LOGARITHMS | LEVEL 3***Tricky – the method has to be found before it can be used.*

1. How many distinct real numbers belong to the following collection?

$$\left\{ \ln(4 - \sqrt{15}), \ln(4 + \sqrt{15}), -\ln(4 - \sqrt{15}), -\ln(4 + \sqrt{15}), \ln\left(\frac{4 + \sqrt{15}}{4 - \sqrt{15}}\right), \ln(31 + 8\sqrt{15}) \right\}$$

(A) 2 (B) 4

(C) 5 (D) 3

2. $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + \cdots + \log_{10}(\log_{1023} 1024)$ simplifies to

- (A) an integer
(B) a composite number
(C) a rational which is not an integer
(D) a prime number

3. The product of all values of x which make the statement $(\log_3 x)(\log_5 9) - \log_x 25 + \log_3 2 = \log_3 54$ true is

- (A) 5 (B) 25
(C) $5\sqrt{5}$ (D) $\sqrt{5}$

4. If x_1 and x_2 satisfy the equation $x^{\log_{10} x} = 100x$, then the value of $x_1 x_2$ equals

- (A) 10 (B) 1000
(C) 100 (D) 1

5. The sum of the squares of the roots of the equation $\log_2(9 - 2^x) = 3 - x$ is

- (A) 5 (B) 10
(C) 9 (D) 3

6. If $\log_{1/8}(\log_{1/4}(\log_{1/2} x)) = \frac{1}{3}$, then x is

- (A) $\frac{1}{\sqrt{2}}$ (B) $\sqrt{2}$
(C) $\frac{1}{2}$ (D) 2

7. Let $\log_b a = 3$ and $\log_b c = -4$. If the value of x satisfying $a^{3x} = c^{x-1}$ is $\frac{p}{q}$ with p, q relatively prime, then $p + q$ is
- (A) 9 (B) 21
(C) 13 (D) 17
8. Let $x = \log 2, y = \log 3$ and $a + bx + cy = [\log 1 + \log(1+3) + \log(1+3+5) + \dots + \log(1+3+5+\dots+19)] - 2[\log 1 + \log 2 + \log 3 + \dots + \log 7]$, where a, b, c are positive integers. Then $2a + 3b + 5c$ equals
- (A) 50 (B) 34
(C) 28 (D) 42
9. The number of values of x satisfying $\log_{10}(2^{x+1} + x - 37) = x(1 - \log_{10} 5)$ is
- (A) 0 (B) 3
(C) 2 (D) 1
10. The solution set of the equation $\log_x 2 \cdot \log_{2x} 2 = \log_{4x} 2$ is
- (A) $\{2^{-2}, 2^2\}$ (B) $\{2^{-\sqrt{2}}, 2^{\sqrt{2}}\}$
(C) $\{\frac{1}{2}, 2\}$ (D) none of these
11. If $|x - 5|^{x^2 - 3x - 10} = 1$, then the sum of all solutions is
- (A) 3 (B) 15
(C) 8 (D) 13
12. The sum of the solutions of the equation $3^{2x+1} + 2 \cdot 3^x \cdot 7^x - 3 \cdot 7^{2x+1} = 0$ is
- (A) 1 (B) -1
(C) 2 (D) 0
13. The values of x satisfying $5^{6x^2} - 2 \cdot 5^{3x^2+5x+2} + 5^{10x+4} = 0$ are
- (A) $2, -\frac{1}{3}$ (B) $-2, \frac{2}{3}$
(C) $-2, -\frac{1}{3}$ (D) $2, \frac{2}{3}$
14. Which of the following is true?
- (A) $3^{150} > 6^{100} > 2^{200}$ (B) $6^{100} > 3^{150} > 2^{200}$
(C) $6^{100} > 2^{200} > 3^{150}$ (D) $3^{150} > 2^{200} > 6^{100}$
15. The product of the solutions of the equation $|\log_{\sqrt{3}} x - 2| - |\log_3 x - 2| = 2$ is

(A) 2

(B) 3

(C) -1

(D) 1

16. If α and β are the solutions of $6^x - 3^{2x-1} - 2^{2x-1} + 6^{x-1} = 0$, then

(A) $\alpha + \beta = 0$ (B) $\alpha + \beta = \frac{1}{1 - \log_2 3}$ (C) $\alpha + \beta = 1$ (D) $\alpha + \beta = \log_2 3$

17. If $2^{2x+1} = 3^y$ and $3^{2y+1} = 2^x$, then $(x - y)$ is equal to

(A) $\log_3 2 - \log_2 3$ (B) $\frac{\log_3 2 - \log_2 3}{3}$ (C) $\frac{\log_3 2 + \log_2 3}{3}$ (D) $\log_3 2 + \log_2 3$

18. The root of the equation $\sqrt{(\log_2 x)^2 - 4\log_2 x + 19} = \log_2(x^2 + 12)$, where $x > 1$ and $\log_2 x, \log_2(x^2 + 12)$ are integers, is

(A) 4

(B) 2

(C) 5

(D) 3

19. The product of all real values of t satisfying $\left| 5^{(\log_5 t)^2} - 25 \right| - 4t^{\log_5 t} = 0$ is

(A) 125

(B) 1

(C) 25

(D) 5

20. Which one of the following is incorrect?

(A) $\log_2 3 > \log_3 11$ (B) If $\frac{\log_8(8/x^2)}{(\log_8 x)^2} = 3$ then $x = \frac{1}{8}$ or 2(C) $\sum_{r=1}^n \log_{n!} r = 1$ for $n \geq 2$ (D) $\sqrt[3]{5^{1/\log_7 5} + \frac{1}{\sqrt{-\log_{10} 0.1}}} = 2$

21. If $\log_2 x = \log_7 8$, $\log_3 y = \log_5 9$ and $\log_5 z = \log_{11} 25$, then $x^{(\log_2 7)^2} + y^{(\log_3 5)^2} + z^{(\log_5 11)^2}$ equals

(A) 389

(B) 289

(C) 489

(D) 589

22. Let α, β, γ be distinct reals with $\frac{1}{(\alpha - \beta)^2} + \frac{1}{(\beta - \gamma)^2} + \frac{1}{(\gamma - \alpha)^2} = 5^{2[\log_5 11 - \log_5 9]}$. Then

$$\left(\frac{1}{\alpha - \beta} + \frac{1}{\beta - \gamma} + \frac{1}{\gamma - \alpha} \right)^2 \text{ is}$$

(A) $\frac{9}{11}$

(B) $\frac{11}{9}$

(C) $\frac{81}{121}$

(D) $\frac{121}{81}$

23. Let p, q be the roots of $x^2 + 2bx + c = 0$ and let $2 \log(\sqrt{y-p} + \sqrt{y-q}) = \log a + \log(y + b + \sqrt{y^2 + 2by + c})$ wherever defined. Then a is

(A) 2

(B) 4

(C) 0

(D) 1

24. If $\frac{1}{p} - \frac{1}{q} = \frac{1}{q} - \frac{1}{r}$, then $\frac{\log(p-r)}{\log(p+r) + \log(p-2q+r)}$ equals (assume all terms are defined)

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{1}{4}$

(D) 4

25. If a, b, c are positive reals, then $\frac{(ab)^{\log \frac{a}{b}} \cdot (bc)^{\log \frac{b}{c}} \cdot (ca)^{\log \frac{c}{a}}}{a^{\log \frac{b}{c}} \cdot b^{\log \frac{c}{a}} \cdot c^{\log \frac{a}{b}}}$ is equal to

(A) -1

(B) 0

(C) 1

(D) none of these